B4 Solutions: Powers, roots and exponential growth

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Rounding: growth factors to 4 decimals unless stated; money to cents; counts of bacteria, people and flies to whole numbers. Intermediate results are kept unrounded in the calculator.

Practice

1.

a) $a^5 \cdot a^3 = a^8$ (add exponents).
b) $\dfrac{x^7}{x^2} = x^5$ (subtract exponents).
c) $(y^3)^4 = y^{12}$ (multiply exponents).
d) $(2p^3)^4 = 2^4(p^3)^4 = 16p^{12}$.
e) $(3a^2b)^2 = 9a^4b^2$, so $(3a^2b)^2 \cdot 2ab^3 = 9 \cdot 2 \cdot a^{4+1}b^{2+3} = 18a^5b^5$.

2.

a) $2^{-3} = \dfrac{1}{2^3} = \dfrac{1}{8}$.
b) $5^0 = 1$.
c) $\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \dfrac{9}{4}$.
d) $(-2)^4 = 16$; $-2^4 = -(2^4) = -16$.
e) $3^{-2} \cdot 3^5 = 3^3 = 27$.
f) $10^{-4} = \dfrac{1}{10\,000} = 0.0001$.

3.

a) $8 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$ (check: $8 \cdot 64 = 512 = 2^9$).
b) $\frac{1}{16} = \frac{1}{2^4} = 2^{-4}$.
c) $\sqrt{2} = 2^{1/2}$.
d) $32^{3/5} = (2^5)^{3/5} = 2^3$ (which is 8).
e) $4^{-3/2} = (2^2)^{-3/2} = 2^{-3}$ (which is $\frac{1}{8}$).
f) $\sqrt[3]{16} = 16^{1/3} = (2^4)^{1/3} = 2^{4/3}$.

4.

a) $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$.
b) $\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$.
c) $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.
d) $\sqrt{18} + \sqrt{8} = 3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2}$.
e) $\sqrt{3} \cdot \sqrt{12} = \sqrt{36} = 6$.
f) $\sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$.
g) $\sqrt{75} - \sqrt{27} = 5\sqrt{3} - 3\sqrt{3} = 2\sqrt{3}$.

5.

a) $\dfrac{6}{\sqrt{3}} = \dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$.
b) $\dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$.

c) $\dfrac{10}{\sqrt{5}} = \dfrac{10\sqrt{5}}{5} = 2\sqrt{5}$.

d) $\dfrac{\sqrt{2}}{\sqrt{6}} = \sqrt{\dfrac{2}{6}} = \dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$.

e) Multiply by $\dfrac{\sqrt{3}+1}{\sqrt{3}+1}$: the denominator is $(\sqrt{3})^2 - 1 = 2$, so $\dfrac{2}{\sqrt{3}-1} = \dfrac{2(\sqrt{3}+1)}{2} = \sqrt{3}+1$.

6.

a) $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$. $16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8$. $32^{-2/5} = \dfrac{1}{(\sqrt[5]{32})^2} = \dfrac{1}{2^2} = \dfrac{1}{4}$. $8^{-1/3} = \dfrac{1}{\sqrt[3]{8}} = \dfrac{1}{2}$. $\left(\frac{1}{9}\right)^{-1/2} = 9^{1/2} = 3$.

b) $\sqrt{x} \cdot x^2 = x^{1/2+2} = x^{5/2}$. $\dfrac{1}{\sqrt[3]{x}} = x^{-1/3}$. $\dfrac{x\sqrt{x}}{x^3} = \dfrac{x^{3/2}}{x^3} = x^{3/2-3} = x^{-3/2}$.

7.

a) $4\,500\,000 = 4.5 \cdot 10^6$; $0.000\,32 = 3.2 \cdot 10^{-4}$.

b) $(3 \cdot 10^5)(4 \cdot 10^{-8}) = 12 \cdot 10^{-3} = 1.2 \cdot 10^{-2}$. $\dfrac{6 \cdot 10^{12}}{1.5 \cdot 10^4} = 4 \cdot 10^8$.

c) Seconds in a year: $365.25 \cdot 24 \cdot 3600 = 31\,557\,600 \approx 3.156 \cdot 10^7$. Distance: $3.0 \cdot 10^8 \cdot 31\,557\,600 = 9.467\ldots \cdot 10^{15}$, so a light-year is about $9.47 \cdot 10^{15}$ m (3 significant figures).

8.

a) $+7\%$: $g = 1.07$. -12%: $g = 0.88$. $+150\%$: $g = 1 + 1.5 = 2.5$.

b) 1.035: $+3.5\%$. 0.94: -6%. 2.5: $+150\%$.

9.

- **A:** differences $4, 4, 4, 4$: linear, $y = 4x + 3$.
- **B:** ratios $2, 2, 2, 2$: exponential, $y = 5 \cdot 2^x$.
- **C:** ratios $\frac{180}{200} = 0.9$, $\frac{162}{180} = 0.9$, $\frac{145.8}{162} = 0.9$, $\frac{131.22}{145.8} = 0.9$: exponential, $y = 200 \cdot 0.9^x$.
- **D:** differences $3, 5, 7, 9$ (not constant), ratios $4, 2.25, \ldots$ (not constant): neither. (It is $y = (x+1)^2$, a quadratic relation.)

10.

a) $g_{\text{day}} = 1.2$. $g_{\text{week}} = 1.2^7 \approx 3.5832$ (an increase of about 258% per week). $g_{\text{hour}} = 1.2^{1/24} \approx 1.0076$ (about $+0.76\%$ per hour; more precisely 1.007626).

b) $1.05^{10} \approx 1.6289$: an increase of about 62.9% in 10 years. Per month: $1.05^{1/12} \approx 1.0041$ (more precisely 1.004074): about 0.41% per month.

11.

a) For even n, $(-x)^n = x^n$, so (x, y) and $(-x, y)$ are both on the graph: $y = x^2$, $y = x^4$ and $y = x^{-2}$ are symmetric in the y-axis. For odd n, $(-x)^n = -x^n$, so with (x, y) also $(-x, -y)$ is on the graph: $y = x^3$ and $y = x^{-1}$ are point-symmetric in the origin. $y = \sqrt{x}$ has neither symmetry (it is not even defined for $x < 0$).

b) $y = x^{-1}$ and $y = x^{-2}$: vertical asymptote $x = 0$ and horizontal asymptote $y = 0$. The others have no asymptotes.

c) At $x = 2$: $x^2 = 4$, $x^3 = 8$, $x^4 = 16$, $x^{-1} = \frac{1}{2}$, $x^{-2} = \frac{1}{4}$, $\sqrt{2} \approx 1.41$. Only $y = x^{-2}$ passes through $(2, \frac{1}{4})$. At $x = -1$: $(-1)^3 = -1$ and $(-1)^{-1} = -1$; $(-1)^2 = (-1)^4 = (-1)^{-2} = 1$; $\sqrt{-1}$ does not exist. So $y = x^3$ and $y = x^{-1}$ pass through $(-1, -1)$.

d) $x^4 = x^2 \cdot x^2$. For $0 < x < 1$ we have $0 < x^2 < 1$, and multiplying the positive number x^2 by a number less than 1 makes it smaller: $x^2 \cdot x^2 < x^2$.

12.

a) The job takes $12 \cdot 10 = 120$ worker-days. 15 workers: $120/15 = 8$ days. 8 workers: $120/8 = 15$ days.

b) $x \cdot y$: $1 \cdot 24 = 2 \cdot 12 = 3 \cdot 8 = 4 \cdot 6 = 6 \cdot 4 = 24$. The product is constant, so x and y are inversely proportional: $y = \dfrac{24}{x}$.

c) Vertical asymptote $x = 0$, horizontal asymptote $y = 0$.

School-exam level

13.

a) $N = 500 \cdot 1.8^5 = 500 \cdot 18.89568 = 9447.84$, so about 9448 bacteria.

b) $g = 1.8 = 1 + 0.80$: 80% per hour.

c) $g_{\text{min}} = 1.8^{1/60} \approx 1.0098$ (more precisely 1.009845).

d) $500 \cdot 1.8^t > 1\,000\,000$ means $1.8^t > 2000$. Table:

t	12	12.9	13.0
N	578 416	981 715	1 041 148

At $t = 12.9$ the number is still below a million, at $t = 13.0$ it is above, so the crossing lies between 12.9 and 13.0 hours. Trying $t = 12.93$ gives $N \approx 999\,179$ and $t = 12.94$ gives $N \approx 1\,005\,070$, so the crossing is at about $t \approx 12.93$. Rounded to one decimal: after about **12.9 hours**. (If the question is read as "at which tenth of an hour is N already above a million", the answer is $t = 13.0$; both readings are fine if explained.)

e) Solve $1.8^T = 2$ with T in hours: $1.8^{1.1} \approx 1.909$ and $1.8^{1.2} \approx 2.025$, so T is between 66 and 72 minutes. Refine in minutes: $1.8^{70/60} \approx 1.9853$, $1.8^{71/60} \approx 2.0048$. The exact value is $T \approx 1.179$ hours ≈ 70.75 minutes, so the doubling time is about **71 minutes**.

14.

a) 24 hours is $24/6 = 4$ half-lives: $400 \cdot \left(\frac{1}{2}\right)^4 = 25$ mg.

b) $g_{6\text{ h}} = 0.5$, so $g_{\text{hour}} = 0.5^{1/6} \approx 0.8909$. The amount decreases by about $1 - 0.8909 = 0.1091$, i.e. 10.9% per hour.

c) $A = 400 \cdot 0.8909^t$, or exactly $A = 400 \cdot 0.5^{t/6}$.

d) Table (with $A = 400 \cdot 0.5^{t/6}$):

t	30	31	32
A	12.50	11.14	9.92

After **32 hours** the amount is below 10 mg for the first time (the exact crossing is at $t \approx 31.9$ h).

15.

a) $2500 \cdot 1.032^{10} \approx 3425.60$: €3425.60.

b) Table of $2500 \cdot 1.032^n$:

n	13	14	15
amount (€)	3765.10	3885.58	4009.92

After **15 years** there is at least €4000.

c) Each year's interest is also calculated on the interest of earlier years, so percentages multiply, not add. The 10-year factor is $1.032^{10} \approx 1.3702$: an increase of about **37.0%**, not 32%.

d) $g_{\text{month}} = 1.032^{1/12} \approx 1.002628$: about **0.263%** per month (not $3.2/12 \approx 0.267\%$).

16.

a) $g = 0.85$. $32\,000 \cdot 0.85^4 = 16\,704.20$: €16 704.20.

b) $32\,000 \cdot 0.85^7 \approx 10\,258.47$ and $32\,000 \cdot 0.85^8 \approx 8\,719.70$. The value is below €10 000 for the first time after **8 years**.

c) $0.85^3 = 0.614125$, so the value has decreased by $1 - 0.614125 = 0.385875$: about **38.6%**.

17.

a) Per 5 years: $g_5 = \dfrac{27\,600}{24\,000} = 1.15$. Per year: $g = 1.15^{1/5} \approx 1.0283$ (more precisely 1.028347), so about **2.83%** per year.

b) $P = 24\,000 \cdot 1.0283^t$ (or exactly $P = 24\,000 \cdot 1.15^{t/5}$).

c) 2030 is $t = 15$, three periods of 5 years: $24\,000 \cdot 1.15^3 = 36\,501$. About **36 500** inhabitants. (With the rounded factor, $24\,000 \cdot 1.0283^{15} \approx 36\,476$; this small difference shows why the exact form is better.)

d) 2010 is one 5-year period before 2015: $24\,000/1.15 \approx 20\,870$ inhabitants. (Equivalently $t = -5$: $24\,000 \cdot 1.15^{-1}$.)

18.

a) Ratios of consecutive counts: $\frac{173}{120} \approx 1.442$, $\frac{249}{173} \approx 1.439$, $\frac{358}{249} \approx 1.438$, $\frac{516}{358} \approx 1.441$. These are almost equal (the time steps are equal), so the growth is approximately exponential.

b) The average ratio is about 1.44, so $g_{2\text{ days}} \approx 1.44$ and $g_{\text{day}} = 1.44^{1/2} = 1.2$.

c) $N = 120 \cdot 1.2^t$ (t in days).

d) $N(3) = 120 \cdot 1.2^3 = 207.36$: about **207** fruit flies.

19.

a) The pattern repeats from $t = 0$ to $t = 80$: period **80 s**. Maximum 42.0 m, minimum 2.0 m. Equilibrium line: $\frac{42+2}{2} = 22$ m. Amplitude: $\frac{42-2}{2} = 20$ m.

b) Diameter $= 42 - 2 = 40$ m (twice the amplitude). The axle is at the height of the equilibrium line: 22 m.

c) $250 = 3 \cdot 80 + 10$, so the height at $t = 250$ equals the height at $t = 10$: **7.9 m**.

d) The graph is a smooth wave between 2 and 42 m, with tops at $t = 40$ and $t = 120$ and bottoms at $t = 0, 80, 160$. In the first turn, h passes 30 m between $t = 20$ ($h = 22$) and $t = 30$ ($h = 36.1$), at about $t \approx 25$ s, and by symmetry comes back down through 30 m at about $t \approx 80 - 25 = 55$ s. So the seat is above 30 m for about **30 s** per turn. (Linear interpolation between the table values gives about 25.7 s and 54.3 s, i.e. about 29 s; with the exact model $h = 22 - 20\cos\left(\frac{2\pi t}{80}\right)$, which you will learn in C4, the answer is 29.5 s. Any estimate between 28 and 31 s is good.)

20.

a) Equilibrium line: $\dfrac{1.2 + (-0.8)}{2} = 0.2$ m above NAP. Amplitude: $\dfrac{1.2 - (-0.8)}{2} = 1.0$ m.

b) Low water is half a period after high water. Half of 12 h 25 min is 6 h 12.5 min. First low water: $03{:}00 + 6{:}12.5 \approx$ **09:12** (09:12 and 30 seconds).

c) $03{:}00 + 12{:}25 =$ **15:25**, and $15{:}25 + 12{:}25 =$ **03:50** the next day.

d) The level passes the equilibrium line a quarter period after high water: $\frac{1}{4} \cdot 12$ h 25 min $= 3$ h 6.25 min. So at about **06:06**.

21.

a) $d = 2$: $I = \dfrac{400}{4} = 100$ lux. $d = 4$: $I = \dfrac{400}{16} = 25$ lux.

b) $I(3d) = \dfrac{400}{(3d)^2} = \dfrac{400}{9d^2} = \dfrac{1}{9} \cdot \dfrac{400}{d^2}$, using $(3d)^2 = 3^2 d^2$. The intensity becomes $\frac{1}{9}$ of what it was.

c) $\dfrac{400}{d^2} = 25$ gives $d^2 = 16$, so $d = 4$ m (distance is positive).

d) The horizontal asymptote is $I = 0$: the further you are from the lamp, the closer the intensity gets to 0, but it never becomes exactly 0.

Challenge

22.

a) $2^{300} = (2^3)^{100} = 8^{100}$ and $3^{200} = (3^2)^{100} = 9^{100}$. Since $8 < 9$, $8^{100} < 9^{100}$: 3^{200} **is bigger**.

b) Raise both (positive) numbers to the 30th power; this keeps the order, because for positive numbers $a < b$ implies $a^{30} < b^{30}$. $\left(10^{1/10}\right)^{30} = 10^3 = 1000$ and $\left(2^{1/3}\right)^{30} = 2^{10} = 1024$. So $\sqrt[3]{2}$ **is bigger**.

c) *Lower bound for* 2^{100}. $2^{100} = (2^{10})^{10} = 1024^{10} = (10^3 \cdot 1.024)^{10} = 10^{30} \cdot 1.024^{10}$. Write $1.024^{10} = (1 + 0.024)(1 + 0.024) \cdots (1 + 0.024)$ (ten brackets). Expanding, choosing 1 from every bracket gives 1; choosing 0.024 from exactly one bracket gives ten terms 0.024; all other terms are positive. So $1.024^{10} > 1 + 10 \cdot 0.024 = 1.24$, and $2^{100} > 1.24 \cdot 10^{30}$. *Upper bound for* 3^{63}. $3^{63} = (3^{21})^3$ and $3^{21} = 10\,460\,353\,203 < 1.05 \cdot 10^{10}$. So $3^{63} < 1.05^3 \cdot 10^{30} = 1.157625 \cdot 10^{30}$. Since $1.157625 < 1.24$: $3^{63} < 1.16 \cdot 10^{30} < 1.24 \cdot 10^{30} < 2^{100}$. 2^{100} **is bigger.** (Check: $2^{100} \approx 1.268 \cdot 10^{30}$ and $3^{63} \approx 1.145 \cdot 10^{30}$. The two are close because $\frac{100}{63} \approx 1.5873$ is only slightly larger than the number x with $2^x = 3$, which is about 1.585.)

23.

a) Last digits of $7^1, 7^2, 7^3, 7^4, 7^5, \ldots$: $7, 9, 3, 1, 7, \ldots$ The last digit of a product only depends on the last digits of the factors, so the pattern repeats with period 4. $2027 = 4 \cdot 506 + 3$, so 7^{2027} ends like $7^3 = 343$: last digit **3**.

b) Powers of 2 end in $2, 4, 8, 6, \ldots$ (period 4), so 2^{2027} ends like 2^3: 8. Powers of 3 end in $3, 9, 7, 1, \ldots$, so 3^{2027} ends like $3^3 = 27$: 7. The sum ends like $8 + 7 = 15$: last digit **5**.

24.

a) *Lemma: if p^2 is a multiple of 3, then p is.* If p is not a multiple of 3, then $p = 3k + 1$ or $p = 3k+2$. Then $p^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$ or $p^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1$; in both cases p^2 leaves remainder 1 on division by 3, so it is not a multiple of 3. *Proof.* Suppose $\sqrt{3} = \frac{p}{q}$ in lowest terms. Then $p^2 = 3q^2$, so p^2 is a multiple of 3, so p is (lemma): $p = 3k$. Then $9k^2 = 3q^2$, so $q^2 = 3k^2$, so q is a multiple of 3 too. Then p and q have common factor 3, contradicting lowest terms. So $\sqrt{3}$ is irrational. ■

b) For $\sqrt{4}$ we would get $p^2 = 4q^2$, and the argument would need "if 4 divides p^2, then 4 divides p". That is false: $p = 2$ (or $p = 6$) has p^2 divisible by 4 but p is not. So the step from p^2 to p breaks. Indeed $\sqrt{4} = \frac{2}{1}$, with $p = 2$, $q = 1$. (The lemma works for 2 and 3 because they are prime; for a prime r, if r divides p^2 then r divides p.)

c) *First, $\sqrt{6}$ is irrational.* Suppose $\sqrt{6} = \frac{p}{q}$ in lowest terms. Then $p^2 = 6q^2$ is even, so p is

even: $p = 2k$. Then $4k^2 = 6q^2$, so $2k^2 = 3q^2$. So $3q^2$ is even; since 3 is odd, q^2 must be even (odd times odd is odd), so q is even. Both even: contradiction. *Now* $\sqrt{2}+\sqrt{3}$. Suppose $\sqrt{2} + \sqrt{3} = r$ with r rational. Squaring: $2 + 2\sqrt{6} + 3 = r^2$, so $\sqrt{6} = \dfrac{r^2 - 5}{2}$. The right side is rational (sums, products and quotients of rationals are rational), but $\sqrt{6}$ is irrational. Contradiction, so $\sqrt{2} + \sqrt{3}$ is irrational. ■

25.

a) $4^x + 4^x + 4^x + 4^x = 4 \cdot 4^x = 4^{x+1} = 2^{2(x+1)} = 2^{2x+2}$. So $2x + 2 = 20$, $x = 9$.
b) $2^x \cdot (2^2)^{x+1} = (2^5)^{x-2}$ gives $2^{x+2x+2} = 2^{5x-10}$, so $3x + 2 = 5x - 10$, $x = 6$. Check: $2^6 \cdot 4^7 = 2^6 \cdot 2^{14} = 2^{20}$ and $32^4 = 2^{20}$.
c) $(3^2)^x = 3 \cdot (3^3)^{x-1}$ gives $3^{2x} = 3^{1+3x-3} = 3^{3x-2}$, so $2x = 3x - 2$, $x = 2$. Check: $9^2 = 81$ and $3 \cdot 27 = 81$. (We used: if $2^m = 2^n$ then $m = n$. This holds because 2^x is increasing, so different exponents give different values.)

26. Idea: $(\sqrt{m} + \sqrt{n})^2 = m + n + 2\sqrt{mn}$. Look for $m + n$ and mn.

a) $3 + 2\sqrt{2}$: need $m + n = 3$, $mn = 2$: $m = 2$, $n = 1$. So $\sqrt{3 + 2\sqrt{2}} = \sqrt{2} + 1$. Check: $(1 + \sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$.
b) $7-4\sqrt{3} = 7-2\sqrt{12}$: need $m+n = 7$, $mn = 12$: $m = 4$, $n = 3$. So $\sqrt{7 - 4\sqrt{3}} = \sqrt{4}-\sqrt{3} = 2 - \sqrt{3}$. We must take $2 - \sqrt{3}$ and not $\sqrt{3} - 2$, because a square root is non-negative and $2 - \sqrt{3} > 0$. Check: $(2 - \sqrt{3})^2 = 4 - 4\sqrt{3} + 3 = 7 - 4\sqrt{3}$.
c) $11 + 6\sqrt{2} = 11 + 2\sqrt{18}$: $m + n = 11$, $mn = 18$: $m = 9$, $n = 2$. So $\sqrt{11 + 6\sqrt{2}} = 3 + \sqrt{2}$. Check: $(3 + \sqrt{2})^2 = 9 + 6\sqrt{2} + 2 = 11 + 6\sqrt{2}$.
d) $5 + 2\sqrt{6}$: $m + n = 5$, $mn = 6$: $m = 3$, $n = 2$. So $\sqrt{5 + 2\sqrt{6}} = \sqrt{3} + \sqrt{2}$. Check: $(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2$.

27. Factor out 2^{2024}:

$$\frac{2^{2027} + 2^{2025}}{2^{2026} - 2^{2024}} = \frac{2^{2024}(2^3 + 2^1)}{2^{2024}(2^2 - 1)} = \frac{8 + 2}{4 - 1} = \frac{10}{3}.$$

28.

a) Doubling time T: solve $(1 + p/100)^T = 2$ by table. Exact values (rounded to 2 decimals) and first whole year with at least doubling:

p	g	doubling time	first whole year	$72/p$
2	1.02	35.00	36	36
6	1.06	11.90	12	12
9	1.09	8.04	9	8
24	1.24	3.22	4	3

(Example check for $p = 2$: $1.02^{35} \approx 1.99989$, just below 2, and $1.02^{36} \approx 2.0399$. So the exact doubling time is 35.003 years, and the first whole year at which the amount has doubled is 36.)

b) Compare the exact doubling time with $72/p$: differences 1.00 ($p = 2$), 0.10 ($p = 6$), 0.04 ($p = 9$), 0.22 ($p = 24$) years. In relative terms: about 2.8% at $p = 2$, 0.9% at $p = 6$, 0.5% at $p = 9$ and 7% at $p = 24$. The rule is best at $p = 9$ (and very good at 6), worst at $p = 24$, and it overestimates noticeably for small rates like 2%.

c) $1.01^{69} \approx 1.9869 < 2$ and $1.01^{70} \approx 2.0068 > 2$, so the doubling time is between 69 and 70 years (exactly about 69.66). The rule of 72 gives 72, which is too big. This suggests a "rule of 70" (or 69) is more accurate for small rates, while 72 is better around 6–10%. (With logarithms in C2 you will see that the exact doubling time is $\dfrac{\ln 2}{\ln(1 + p/100)}$, and that for small p this is close to $\dfrac{69.3}{p}$.)